

CHE 363 Separation Processes and Mass Transfer

Spring 2022

This course covers the general principles of separation by equilibrium and rate processes. Topics include staged cascades and applications to distillation, absorption, adsorption, and membrane processes. Phase equilibria and the role of diffusion are also covered.

Instructor: Thomas Edison

TA:

Office: CPE 4.458 (1-4438)

Office hours:

Office hours

Email: tedison@che.utexas.edu

Lectures: TTh 11:00 – 12:15 at CPE 2.220

Please be punctual. Attendance is not compulsory, but is strongly recommended.

Recommended Calculator (**TI-Nspire CX CAS**). Bring your calculator to class.

Should be able to solve system of linear and non-linear algebraic equations.

Should be able to do numerical integration.

Recitation: F 1:00 – 2:00 at CPE 2.206

Optional Books: “Separation Process Principles”, 2nd ed. or 3rd ed., J.D Seader and Henley, John Wiley & Sons, 2006.

“Separation Process Engineering: Includes Mass Transfer Analysis”, 3rd ed., Phillip C. Wankat, Prentice Hall, 2012.

“Using ASPEN PLUS in Thermodynamics Instruction: A step-by step Guide”, Stanley I Sandler, John Wiley & Sons, 2015.

Final Grade:	Homework	15 %
	Quiz	10 %
	Aspen HW and Project	10 %
	Exams (3 out of 4)	65 %

Final letter grade will be given based on Class average. The class average for this course is B-. There are two requirements for a passing grade C-: You should have an exam grade of 60 points and have a course average that will be determined from your class average and standard deviation. This will be determined after the third exam.

Homework:

Homework is weekly. Late homework will not be accepted. Homework must be handed in individually, but you are encouraged to work in groups to discuss and solve the problems. Homework problems are graded with an emphasis on effort and setting up the problem, and only selected problems within each assignment will be graded.

Quizzes:

Every week there will be an online quiz. Each quiz may have two to three short problems. The quiz problems will highlight the important concepts covered in class lectures and also will illustrate the finer parts of homework problems. You have to work alone and may not take outside help. Academic integrity is expected.

Recitation: During recitation you will be introduced to ASPEN. Attendance and Laptop are mandatory. You can download ASPEN software to your computer. Aspen lab quiz will check your proficiency of ASPEN.

Exams:

There will be three exams during the semester and a final exam. Exams are closed book and closed notes. All the exams are waited equally. Best of three exams will be counted toward the final grade.

Exam – I	Feb 10 th	(7:00 – 10:00 PM)	CPE 2.218
Exam – II	March 31 th	(7:00 – 10:00 PM)	CPE 2.218
Exam – III	May 5 th	(7:00 – 10:00 PM)	CPE 2.218
Final Exam	May 12 th	(9:00 – 12:00 Noon)	

(You have one week after the exam is returned to you to request for re-grading. Re-grading requests should be submitted in writing at the end of that lecture period.)

The University of Austin provides upon request appropriate academic accommodations for qualified students with disabilities. For more information, contact the Office of the Dean of Students at 471-6259, 471-6441 TTY.

Absence: No excuse for missed tests, exams, or quizzes will be accepted. Lowest quiz grade will be dropped.

Academic Integrity:

The University has a strict policy on academic integrity. Any form of plagiarism or academic dishonesty will NOT be tolerated in ChE 363. If you have any questions, please consult the websites below:

http://deanofstudents.utexas.edu/sjs/acint_student.php

http://www.utexas.edu/courses/clubmed/plgrism_cit.html

In particular, students are obliged to follow the Student Honor Code that states “As a student of The University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity.”

Failure to comply with the University’s policy will result in a “zero” in the assignment and an “F” for CHE 363.

Note: During the lecture period you may sit where you feel comfortable. If any groups of students constantly talk and disturb others during lectures, then the instructor has the right to assign seating arrangement for the benefit of the class.

Knowledge, Abilities, and Skills Students Should Have Entering This Course:

1. Ability to use and solve mass and energy balances (ChE 317)
2. Familiarity with vapor-liquid equilibrium and liquid-liquid equilibrium (ChE 322)
3. Ability to use equations of state and activity coefficient models to calculate fugacities and activity coefficients (ChE 322)
4. Ability to use Raoult’s law and familiarity with partial pressures (ChE 317).
5. Ability to calculate dew points, bubble points and the ability to perform flash calculations
6. Ability to understand and use Fick’s law and diffusion coefficients (ChE 319)
7. Familiarity with analogies between correlations for heat and mass transfer coefficients. (ChE 319)

Knowledge, Abilities, and Skills Students Should Gain From This Course:

Thermodynamics:

1. Understand and select thermodynamic property sets for process simulators
2. Calculate bubblepoint, dewpoint, and flash calculations with simulator and spreadsheet
3. Know and use definitions of K, relative volatility, and activity coefficients
4. Know and use isotherms representations of adsorption solid / fluid equilibrium data

Mass Transfer

1. Use heat and mass transfer analogies and dimensionless groups to estimate mass transfer coefficients
2. Know and use film theory, penetration theory, surface renewal theory
3. Understand unicomponent and equimolar diffusion
4. Understand and use two film mass transfer theory

Distillation, Absorption and Stripping

1. Understand significance of multistage, countercurrent, co-current, well mixed, and crossflow geometries
2. Understand cascades and reflux
3. Understand the concept of key components
4. Understand and estimate separation factors, number of stages, and minimum reflux
5. Find optimum reflux, feed point, preheating, and stages using graphical methods and simulators
6. Understand and use tray efficiencies
7. Select operating pressure

Gas/Liquid Contactors

1. Estimate capacity; understand flooding and entrainment
2. Select contactor type (trays, packing, sprays, bubble column)
3. Estimate pressure drop and energy requirements
4. Estimate mass transfer coefficients, and tray efficiencies

Membrane Separations

1. Understand analogies to heat exchanger design
2. Estimate mass transfer coefficients for membrane modules
3. Understand permeability

Adsorption

1. Understand equilibrium capacity limits
2. Calculate breakthrough behavior
3. Develop mathematical models for adsorption processes

Process Simulation Models

1. Understand structure of process simulation models
2. Know how to construct a flowsheet using a simulator
3. Understand and use models for distillation, absorption, extraction, and membrane separations